

Complete Developer Roadmap

Roadmap based on current skills: React, Learning Django, and interest in System Design.

PHASE 1 — Python Fundamentals

- Variables
- Data types
- Conditions
- Loops
- Functions
- OOP concepts
- File handling
- Exception handling
- Modules & packages
- Virtual environment

PHASE 2 — Django Basics

- Django project structure
- Apps
- URLs
- Views
- Templates
- Static files
- Forms
- Admin panel
- Models
- Migrations

PHASE 3 — Database Fundamentals

- SQL basics
- PostgreSQL basics
- CRUD operations
- Relationships
- One-to-One
- One-to-Many

- Many-to-Many
- Joins
- Indexing basics
- Query optimization basics

PHASE 4 — Django REST Framework

- APIs
- Serializers
- API Views
- ViewSets
- Routers
- CRUD APIs
- Authentication
- JWT
- Permissions
- Pagination
- Filtering
- File upload APIs

PHASE 5 — Authentication & Security

- JWT Authentication
- Session authentication
- Password hashing
- CORS
- CSRF
- Environment variables
- Secure APIs
- Role-based access

PHASE 6 — Git & GitHub

- git init
- git add
- git commit
- git push
- branches
- merge

- pull requests
- resolving conflicts
- .gitignore

PHASE 7 — Build Real Projects

- Blog API
- Authentication system
- Property management app
- Booking system
- Chat application
- Subscription/payment system

PHASE 8 — System Design Basics

- Client vs Server
- HTTP/HTTPS
- DNS
- APIs
- Cookies & Sessions
- Monolith
- Microservices basics
- Vertical Scaling
- Horizontal Scaling
- Load Balancer
- Caching
- CDN
- Database optimization
- Replication
- Failover
- High Availability

PHASE 9 — Linux Basics

- Linux file system
- Terminal commands
- Permissions
- SSH
- Processes

- Package installation
- Logs
- Environment variables

PHASE 10 — Deployment Basics

- Hosting basics
- Gunicorn
- Static/media files
- Production settings
- Environment variables
- Domain basics

PHASE 11 — Nginx

- Reverse proxy
- Serving static files
- Request forwarding
- SSL basics

PHASE 12 — Docker

- Containers
- Images
- Dockerfile
- docker-compose
- Container networking
- Running Django in Docker

PHASE 13 — Cloud Basics

- AWS
- EC2
- S3
- RDS
- IAM basics
- Deployment basics

PHASE 14 — Advanced Backend

- Redis
- Celery
- Background tasks
- WebSockets
- Rate limiting
- Message queues
- Email services

PHASE 15 — Advanced System Design

- Distributed systems
- CAP theorem
- Event-driven architecture
- Kafka basics
- Sharding
- Database replication
- Scalability patterns

PHASE 16 — CI/CD

- Automated deployment
- GitHub Actions
- Build pipelines
- Testing basics

PHASE 17 — Optional Future Learning

- Kubernetes
- Data Engineering
- AI Integration
- Microservices
- DevOps
- Cybersecurity

Best Learning Strategy: Learn → Build Projects → Practice → Deploy → Repeat